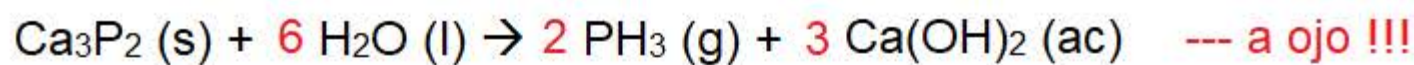


## ESTEQUIOMETRIA



$$\begin{array}{ccccccc} 1 \text{ mol} & 6 \text{ mol} & 2 \text{ mol} & 3 \text{ mol} & & & \\ 182.18 \text{ g} & + 108.00 \text{ g} & = 67.94 \text{ g} & + 222.24 \text{ g} & = & 290.18 \text{ g} & \end{array}$$

$$\text{Mr} (\text{Ca}_3\text{P}_2) = 40.08 \times 3 + 30.97 \times 2$$

$$120.24 + 61.94 = 182.18 \quad (\times 1)$$

$$\text{Mr} (\text{PH}_3) = 30.97 + 3 \times 1.00 = 33.97 \quad (\times 2)$$

$$\text{Mr} [\text{Ca}(\text{OH})_2] = 40.08 + 2 \times 16.00 + 2 \times 1.00$$

$$40.08 + 32.00 + 2.00 = 74.08 \quad (\times 3)$$

b) 240 g fosfuro (90 % P)  $\Rightarrow$  216 g  $\text{Ca}_3\text{P}_2$  puros

180 mL agua (d = 1 g/mL)  $\Rightarrow$  180 g agua

¿RL? 182.18 g fosf ---- 108 g agua

216 g fosf ---- x = 128.05 g agua --- necesito  
180 g agua --- tengo } sobra  $\text{H}_2\text{O}$

RL = fosfuro

182.18 g fosf --- 2 mol  $\text{PH}_3$

216 g fosf --- x = 2.371 mol  $\text{PH}_3$  (si fuera 100 % R)

$\hookrightarrow$  con 75 % R  $\Rightarrow$  1.778 mol  $\text{PH}_3$

$$n = 1.778 \text{ mol}$$

$$T = 27^\circ\text{C} = 300 \text{ K}$$

$$V = 20 \text{ L}$$

$$\left. \begin{array}{l} n = 1.778 \text{ mol} \\ T = 27^\circ\text{C} = 300 \text{ K} \\ V = 20 \text{ L} \end{array} \right\} P = \frac{n \cdot R \cdot T}{V} \Rightarrow \boxed{P = 2.187 \text{ atm}}$$

2 mol  $\text{PH}_3$  ---  $\overset{3 \text{ mol}}{222.24 \text{ g Ca}(\text{OH})_2}$

1.778 mol  $\text{PH}_3$  --- x =  $\boxed{197.57 \text{ g Ca}(\text{OH})_2}$

ya viene con 75 % R

c) 0.60 mol  $\text{PH}_3$  --- al 75 % R

$$\hookrightarrow 0.60 \cdot \frac{100}{75} = 0.8 \text{ mol } \text{PH}_3 \text{ --- si fuera 100 \% R}$$

2 mol  $\text{PH}_3$  --- 182.18 g fosfuro

0.8 mol  $\text{PH}_3$  --- x = 72.782 g fosfuro (puro)

$$\hookrightarrow 72.872 \text{ g} \cdot \frac{100}{90} = 80.969 \text{ g fosfuro (imp)}$$

$\boxed{80.97 \text{ g fosfuro}}$

2 mol  $\text{PH}_3$  --- 108 g agua

0.8 mol  $\text{PH}_3$  --- x = 43.2 g agua  $\Rightarrow$   $\boxed{43.2 \text{ mL agua}}$

## GASES

i)  $X_i = n_i / n_{\text{total}}$

a)  $5.60 \text{ mol CH}_4$   $x \text{ CH}_4 = 5.60 / 8.00 = 0.70$   
 $2.40 \text{ mol C}_2\text{H}_6$   $x \text{ C}_2\text{H}_6 = 2.40 / 8.00 = 0.30$  ) **suman 1**

8.00 mol totales

$P_i = X_i \cdot P_{\text{total}}$

$P \text{ CH}_4 = 0.70 \cdot 1.52 \text{ atm} = 1.064 \text{ atm}$  )

$P \text{ C}_2\text{H}_6 = 0.30 \cdot 1.52 \text{ atm} = 0.456 \text{ atm}$  )

**suman 1.52 atm**

ii) si se agrega  $\text{N}_2$  manteniendo T y V constantes:

$\uparrow n_{\text{totales}} \Rightarrow \uparrow P_{\text{total}}$

$\downarrow X_i$  porque son los mismos  $n_i$  divididos por mas  $n_{\text{tot}}$

$P_i$  no cambia (xq solo depende de  $n_i$ )

$\uparrow$  densidad porque hay mayor masa de gas en el mismo V

b)  $m = 320 \text{ g}$

$P = 1.222 \text{ atm}$

$T = 25^\circ\text{C} = 298 \text{ K}$

$d = 1.60 \text{ g/L}$

$V = \frac{m}{d} = \frac{320 \text{ g}}{1.6 \text{ g/L}} = \underline{200 \text{ L}}$

$n = \frac{P \cdot V}{R \cdot T} = \frac{1.222 \cdot 200}{0.082 \cdot 298} = 10.0 \text{ moles}$

$M_r = \frac{m}{n} = \frac{320 \text{ g}}{10 \text{ mol}} = \underline{32.0 \text{ g/mol}}$

ii) aumenta T a V = cte

$P_1 = 1.222 \text{ atm}$

$T_1 = 25^\circ\text{C} = 298 \text{ K}$

$P_2 = ?$

$T_2 = 250^\circ\text{C} = 523 \text{ K}$

$\frac{P_1 \cdot \cancel{V_1}}{T_1} = \frac{P_2 \cdot \cancel{V_2}}{T_2}$

$\frac{1.222 \text{ atm}}{298 \text{ K}} = \frac{P_2}{523 \text{ K}} \Rightarrow \underline{P_2 = 2.145 \text{ atm}}$

ii) se abre la valvula

$P_{\text{final}} = 2.00 \text{ atm}$  )

$T = 523 \text{ K}$

$V = 200 \text{ L}$

$n = \frac{P \cdot V}{R \cdot T} = \frac{2 \cdot 200}{0.082 \cdot 523} = \underline{9.327 \text{ moles}}$   
(quedan)

inicialmente --- 10 moles

quedan ----- 9.327 moles )

$10 - 9.327 = \underline{0.673 \text{ moles}}$  se escapan  
del cilindro



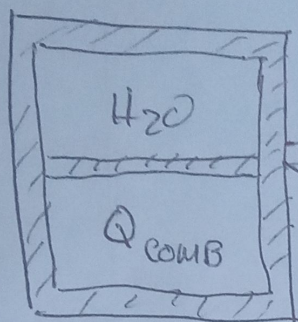
### PROBLEMA 3

$$T_i = 10^\circ\text{C}$$

$$T_f = 25^\circ\text{C}$$

/// PAREDES DEL MOTOR (Al)

$$V_{H_2O} = 10\text{ L}$$



$Q=0$   
SUPERFICIE

SISTEMA

ADIABATICO

SOLO EL 60% DEL CALOR

PRODUCIDO POR LA COMBUSTION  
CALIENTA EL MOTOR (Al + H<sub>2</sub>O)

$$Q_{H_2O} + Q_{Al} + Q_{comb} = 0$$

$$m_{H_2O} \cdot c_{pH_2O} \Delta T + m_{Al} \cdot c_{pAl} \cdot \Delta T + Q_{comb} = 0$$

~~$10\text{ kg} \cdot 4,18$~~

$$10000\text{ g} \cdot 4,18 \frac{\text{J}}{\text{g} \cdot ^\circ\text{C}} (25^\circ\text{C} - 10^\circ\text{C}) + 300000\text{ g} \cdot 0,92 \frac{\text{J}}{\text{g} \cdot ^\circ\text{C}} (25^\circ\text{C} - 10^\circ\text{C}) + Q_{comb} = 0$$

$$627\text{ KJ} + 4140\text{ KJ} + Q_{comb} = 0$$

$$Q_{comb} = -4767\text{ KJ}$$

SI SOLO EL 60% DEL CALOR ES USADO PARA  
CALENTAR:

$$4767\text{ KJ} - 60\% \quad x = 7945\text{ KJ}$$

$$x - 100\%$$

$$A \quad p = c \cdot H \cdot e \quad Q = n \Delta H_{comb}$$

$$7945\text{ KJ} = n_{OCTANO} \cdot \Delta H_{comb} \Rightarrow n_{OCTANO} = 1,566\text{ mol}$$

$$V_{OCTANO} = 1,566\text{ mol} \cdot \frac{114\text{ gr}}{\text{mol}} \cdot \frac{1\text{ ml}}{0,7\text{ g}} = 255\text{ ml}$$



P3

$$b) \Delta H_r^\circ = -5072,1 \frac{\text{kJ}}{\text{mol}} \quad \text{P}^\circ$$

$$\Delta H_r^\circ = 9\Delta H_f^\circ(\text{H}_2\text{O}) + 8\Delta H_f^\circ(\text{CO}_2) - 12,5\Delta H_f^\circ(\text{O}_2) - \Delta H_f^\circ(\text{C}_8\text{H}_{18})$$

$$\Delta H_f^\circ(\text{C}_8\text{H}_{18}) = 9 \cdot (241,8 \frac{\text{kJ}}{\text{mol}}) + 8 \cdot (-393,5 \frac{\text{kJ}}{\text{mol}}) - (-5072,1 \frac{\text{kJ}}{\text{mol}})$$

$$\Delta H_f^\circ(\text{C}_8\text{H}_{18}) = -252,1 \frac{\text{kJ}}{\text{mol}}$$

$$c) \Delta H_r^\circ = D_{\text{REACT}} - D_{\text{PRODUC.}}$$

$$= 7D_{\text{C-C}} + 18D_{\text{C-H}} + 12,5D_{\text{O=O}} - 8 \cdot 2 \cdot D_{\text{C=O}} - 9 \cdot 2 \cdot D_{\text{O-H}}$$

$$= 7 \cdot 350 \frac{\text{kJ}}{\text{mol}} + 18 \cdot 410 \frac{\text{kJ}}{\text{mol}} + 12,5 \cdot 498 \frac{\text{kJ}}{\text{mol}} - 16 \cdot 732 \frac{\text{kJ}}{\text{mol}} - 18 \cdot 460 \frac{\text{kJ}}{\text{mol}}$$

$$\Delta H_r^\circ = -3937 \frac{\text{kJ}}{\text{mol}}$$

d) (EXTRA)

$$C_{PAI} = 3R = 3 \cdot 8,314 \frac{\text{J}}{\text{mol} \cdot ^\circ\text{C}} \cdot \frac{1 \text{ mol}}{27 \text{ g}} = 0,924 \frac{\text{J}}{\text{g} \cdot ^\circ\text{C}}$$



## PROBLEMA 4

a)  $\Delta S > 0$ , PORQUE EN PRODUCTOS HAY MAS MOLES DE GAS QUE EN REACTIVOS. LA ENTROPIA DE LOS GASES ES MAYOR A LA DE LOS LIQUIDOS.

$$\begin{aligned} \text{b) } \Delta S^\circ &= 2S^\circ_{\text{H}_2\text{O}} + 2S^\circ_{\text{N}_2\text{O}} - S^\circ_{\text{N}_2\text{O}_4} - S^\circ_{\text{N}_2\text{H}_4} \\ &= (2 \cdot 188,8 + 2 \cdot 220 - 304,4 - 121,2) \text{ J/mol}\cdot\text{K} \\ &= 392 \text{ J/mol}\cdot\text{K} \end{aligned}$$

$$\begin{aligned} \text{c) } \Delta G^\circ_r &= \Delta G^\circ_{\text{PROD}} - \Delta G^\circ_{\text{REACT}} = 2 \cdot \Delta G^\circ_{f, \text{H}_2\text{O}} + 2 \Delta G^\circ_{f, \text{N}_2\text{O}} - \Delta G^\circ_{f, \text{N}_2\text{O}_4} - \Delta G^\circ_{f, \text{N}_2\text{H}_4} \\ \Delta G^\circ_r &= ~~498~~ - 498,9 \text{ kJ/mol} \end{aligned}$$

$$\Delta G^\circ_{f, \text{N}_2\text{H}_4} = 498,9 \frac{\text{kJ}}{\text{mol}} + 2 \cdot (-228,6) + 2 \cdot (103,7) - 99,8$$

$$\Delta G^\circ_{f, \text{N}_2\text{H}_4} = 149,3 \frac{\text{kJ}}{\text{mol}}$$

$$\text{d) } \Delta G = \Delta H - T\Delta S \Rightarrow \Delta H = \Delta G + T\Delta S$$

$$\Delta H = -498,9 \frac{\text{kJ}}{\text{mol}} + 298 \text{ K} \cdot 393 \times 10^{-3} \frac{\text{kJ}}{\text{mol}\cdot\text{K}} = -381,79 \frac{\text{kJ}}{\text{mol}}$$

e)  $\Delta H < 0$ , RXN EXOTERMICA, LIBERA CALOR

f) ES ESPONTANEA:  $\Delta G^\circ_r < 0$

$$\text{g) } \Delta G = \Delta H - T\Delta S$$

COMO  $\Delta S > 0 \rightarrow -T\Delta S$  SERA NEGATIVO A CUALQUIER TEMP.

COMO  $\Delta H < 0 \Rightarrow \Delta G < 0$  A CUALQUIER TEMP!